

Functions and Pigeonhole principle

Example

Ex. A function f is defined $f: \mathbb{R}^+ \rightarrow \mathbb{N}$
 $f(x) = \lfloor x \rfloor$ (the largest int $\leq x$).
Is f , one to one, onto or both?

* Can assume co domain $\mathbb{N} \cup \{0\}$

Solution

fun one to one

$$f(2.5) = \lfloor 2.5 \rfloor = 2$$

$$f(2.6) = \lfloor 2.6 \rfloor = 2$$

$2.5 \neq 2.6$ but still $f(2.5) = f(2.6)$
 f is not one to one.

~ $\forall y \in \mathbb{N}$, we have $\exists x \in \mathbb{R}^+$ such
that $f(x) = y$.

~ So, it is onto

Example

Ex. Check whether the function is one to one or onto.

$f: \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z} \times \mathbb{Z}$, defined by
 $f(a, b) = (a+b, a-b)$. ($\mathbb{Z}$ is Set of int)

for one to one

assume

$$f(a, b) = f(c, d)$$

$$\Rightarrow (a+b, a-b) = (c+d, c-d)$$

$$\Rightarrow \begin{array}{ll} a+b = c+d & \text{--- (1)} \\ a-b = c-d & \text{--- (2)} \end{array}$$

$$1 + 2, \quad a = c$$

$$1 - 2, \quad b = d$$

So, function is one to one

- If for every element of Co-domain there is at least one element in X such that $f(x) = y$.

Need to show

$$\forall (x, y) \in I \times Z$$

$$\exists (a, b) \in Z \times Z \text{ such that}$$

$$f(a, b) = (x, y)$$

$$\text{let, } f(a, b) = (x, y)$$

$$(a+b, a-b) = (x, y)$$

$$a+b = x$$

$$a-b = y$$

$$a = \frac{x+y}{2}, \quad b = \frac{x-y}{2}$$

So, a and b does not always belong to $\mathbb{Z}$.

So, f is not onto / Surjective

Formula for number of Functions

Number of Functions

- If a set A has m elements and set B has n elements, then the number of functions possible from A to B is n^m .
- For example, if set $A = \{3, 4, 5\}$, $B = \{a, b\}$.
 - The total number of possible functions from A to B = $2^3 = 8$

Number of one to one functions

$$f: A \rightarrow B$$

Cardinality of $A : m$

Cardinality of $B : n$

Total No. of one to one fun.

$$= \begin{cases} n P_m, & n \geq m \\ 0 & n < m \end{cases}$$

Number of onto functions

$$f: A \rightarrow B$$

Cardinality of $A : m$

Cardinality of $B : n$

Total No. of onto fun

$$= \sum_{r=1}^n (-1)^{n-r} \cdot n C_r \cdot n^m \quad \text{if } m \geq n$$

$$= 0, \quad m < n$$

Example-1

- Find the number of injective function from A to B where $A = \{ a, b, c \}$ and $B = \{ 1, 2, 3, 4, 5 \}$

$$f: A \rightarrow B$$

Cardinality of $A : m$

Cardinality of $B : n$

$$m = 3, \quad n = 5$$

No. of functions

$$= {}^n P_m = \frac{n!}{(n-m)!} = \frac{5!}{2!} = 60$$

Example-2

- Count the number of surjective functions from A to B where $A=\{1,2,3,4\}$ and $B=\{a,b\}$.

$$m=4, n=2$$

$$\text{No. of fun} = \sum_{r=1}^n (-1)^{n-r} n C_r r^m$$

$$= \sum_{r=1}^2 (-1)^{2-r} 2 C_r r^4$$

$$= (-1)^{2-1} \cdot 2 C_1 \cdot (1)^4 + (-1)^{2-2} \cdot 2 C_2 (2)^4$$

$$= (-1) \cdot \frac{2!}{1! (2-1)!} (1) + (1) \cdot \frac{2!}{2! (2-2)!} \times 16$$

$$= (-1) \times 2 + 16 = 14$$

Pigeonhole Principle

- The pigeonhole principle states that if n items are put into m containers, with $n > m$, then at least one container must contain more than one item.
- Also known as shoe box argument.

- If there are many pigeons and a few pigeonholes then there must be some pigeonhole occupied by two or more pigeons.
- Let D and R be finite sets if $|D| > |R|$, then for any function f from $D \rightarrow R$, there exists $d_1, d_2 \in D$ such that $f(d_1) = f(d_2)$

- For any function f from $D \rightarrow R$ there exists i elements $d_1, d_2, \dots, d_i$ in D , $i = \lceil |D|/|R| \rceil$ such that $f(d_1) = f(d_2) = \dots = f(d_i)$.

Example-1

- If 50 bicycles are colored with 7 colors then at least how many bicycles have the same color ?

$n = \text{bicycles} = 50$ (pigeon)

$m = \text{Colors} = 7$ (pigeonholes)

$$\lceil \frac{n}{m} \rceil = \lceil \frac{50}{7} \rceil = 8$$

✓	1	B	G	y	o	R
✓1	2	3	4	5	6	7
✓8	9	10	11	12	13	14
✓						21
✓						28
✓						35
✓						42
✓						49
✓50						

Example-2

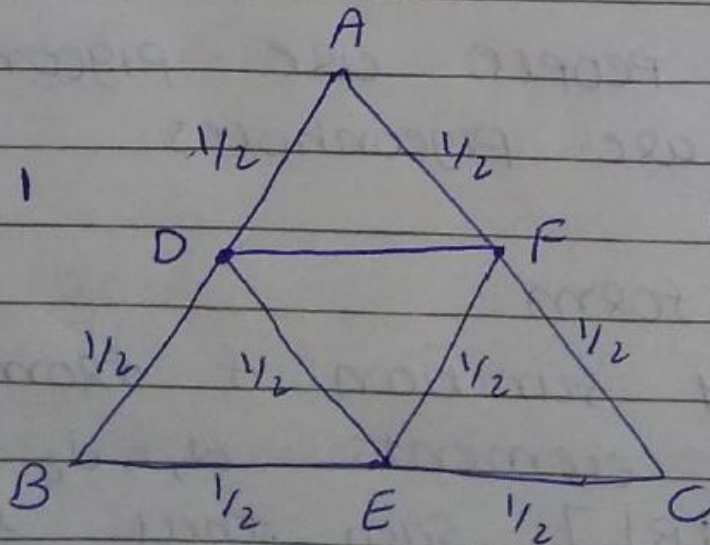
- Let T be an equilateral triangle whose sides are of length 1 unit. Show that if any 5 points are chosen lying on or inside triangle, then two of them must be no more than 0.5 unit apart.

Ans: Let T be the equilateral triangle
 $\triangle ABC$

we are given that $AB = BC = CA = 1$ unit

→ Let D, E and F be the mid points of
the sides AB, BC, CA respectively

→ Join DE, EF and FD



NOW we have

$$AD = DB = BE = EC = AF = FC = DF = DE = EF = \frac{1}{2} \text{ unit}$$

NOW triangle ABC is divided into 4 equilateral triangle ADF, DBE, EFC and DEF, each possessing side of length $\frac{1}{2}$ units

NOW, each of the 5 points Chosen must belong to one of these triangles (Pigeonholes)

→ we consider the points as the pigeons, since there are only 4 triangles and any two points chosen from a triangle out of these 4 are not more than $\frac{1}{2}$ unit apart

∴ According to pigeonhole principle if 5 points are chosen then two of them must be no more than $\frac{1}{2}$ unit apart (because two points must belong to a single triangle)

Example-3

A chess player wants to prepare for a **championship match by playing some practice games in 77** days. She wants to play at least one game a day but not more than 132 games altogether. Show that no matter how she schedules the games, there is a period of consecutive days during which she plays exactly 21 games.

Solution

Example 4.4 A chess player wants to prepare for a championship match by playing some practice games in 77 days. She wants to play at least one game a day but no more than 132 games altogether. We show now that no matter how she schedules the games there is a period of consecutive days within which she plays *exactly* 21 games. Let a_i denote the total number of games she plays up through the i th day. Clearly, the sequence $a_1, a_2, \dots, a_{77}$ is a monotonically increasing sequence, with $a_1 \geq 1$ and $a_{77} \leq 132$. Let us compute the sequence $a_1 + 21, a_2 + 21, \dots, a_{77} + 21$, which again is a monotonically increasing sequence with $a_{77} + 21 \leq 153$. Since the values of the 154 numbers $a_1, a_2, \dots, a_{77}, a_1 + 21, \dots, a_{77} + 21$ range from 1 to 153, two of them must be the same. Moreover, because both the sequence $a_1, a_2, \dots, a_{77}$ and the sequence $a_1 + 21, a_2 + 21, \dots, a_{77} + 21$ are monotonically increasing, we have $a_i = a_j + 21$ for some a_i and a_j . $\square$